(i)

$$x + 3\sqrt{x} - \frac{1}{2} = 0$$

$$y^2 + 3y - \frac{1}{2} = 0 \implies y = \frac{-3 \pm \sqrt{11}}{2}$$

$$\because y \geq 0$$

$$\therefore x = y^2 = \left(\frac{\sqrt{11} - 3}{2}\right)^2 = \frac{10 - 3\sqrt{11}}{2}$$

(ii)(a) Let $y = \sqrt{x+2},\ y \geq 0$

$$x + 10\sqrt{x+2} - 22 = 0$$
$$y^2 - 2 + 10y - 22 = 0$$
$$y^2 + 10y - 24 = 0$$
$$(y + 12)(y - 2) = 0$$

$$\because y \geq 0$$
$$x + 2 = y = 4 \implies x = 2$$

(ii)(b) Let $y = x^2 - 4x,\ u = \sqrt{2y-3},\ y = \dfrac{u^2+3}{2}$

$$x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$$
$$y + \sqrt{2y - 3} - 9 = 0$$
$$\frac{u^2 + 3}{2} + u - 9 = 0$$
$$(u + 5)(u - 3) = 0 \implies u = 3 \implies y = 6$$

$$x^2 - 4x = 6 \implies x = 2 \pm \sqrt{10}$$

This solution was submitted by **Jon**
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